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Quantum color image median filtering in the spatial domain: theory and experiment

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Abstract

Median filters are widely used to smooth data in image processing tasks and eliminate noise with extremely large or small magnitude. Thus, this paper proposes a median filtering method for quantum color images in the spatial domain, with the corresponding quantum circuit simulated on IBM the Quantum Experience (IBM Q) platform. In our technique, first, we improve the quantum color image storage module and adopt nine quantum color images to represent the core image (image to be processed) and its 8-neighborhoods. Then, a quantum comparator and a quantum control swap operation with low time complexity are employed to design the quantum median calculation module. Finally, we design the quantum circuit for color image median filtering, simulated on the IBM Q platform. Compared with existing methods, the proposed method's time complexity is significantly reduced from $O(q^2)$ to $O(q)$ and the auxiliary qubit is reduced from $O(q)$ to $O(1)$. The developed scheme provides the framework for encoding the neighborhood information of pixels in supplementary images and reduces the auxiliary qubits and time complexity, demonstrating the potential of quantum image processing for highly efficient image and video processing tasks in the big data era.

Keywords Quantum color image processing · Median filtering algorithm · Quantum image simulation

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1 Introduction

Currently, image processing is a rapidly developing technology, with the quantum computing and image processing combination gaining significant research attention. Processing images with quantum computers starts with the classic encoding of images into quantum states, while, later, many quantum image representation models have been proposed. In 2003, Venegas-Andraca proposed for the first time the image representation model of Qubit Lattice-based on a Qubit array [1]. Spurred by this research breakthrough, several works extended it and suggested various models such as the flexible representation of quantum images (FQRI) [2], the novel enhanced quantum representation (NEQR) [3], an n-qubit normal arbitrary superposition state (NASS) [4], the improved novel enhanced quantum representation (INEQR) [5], the generalized quantum image representation (GQIR) [6], the novel quantum representation of color digital images (NCQI) [7], an optimized quantum representation for color digital images (OCQR) [8] model, and a quantum hue, saturation, and lightness color model (QHSL) [9]. Compared with other quantum color image representation models, OCQR can flexibly manipulate pixels and requires fewer qubits. Therefore, this paper adopts the OCQR model to represent quantum color imagery.

Based on these quantum image representation models, various quantum image processing algorithms have been proposed, such as geometric transformation [10–12], image scaling [5, 6, 13], image morphology processing [14, 15], image segmentation [16, 17], image encryption [18–21], image scrambling [22], image watermarking [23], image compression [24] and image filtering [25–33]. Considering the frequency domain filtering, in [25], the authors develop the image frequency domain filtering algorithm that is based on the quantum Fourier transform, which has already been extended to facilitate color images [26]. Both methods use the quantum Fourier transform and quantum Oracle to achieve highly efficient frequency domain filtering. However, the functional module represented by quantum Oracle does not involve a specific quantum circuit, and thus, it is impossible to verify whether the algorithm is feasible. Therefore, this kind of algorithm has certain limitations. Regarding the space domain, Yuan et al. [27, 28] suggest quantum image filtering based on an adder operation and multiplier operation, respectively, while other works utilize mean filtering [29], median filtering [30], improved median filtering [31], and quantum midpoint filtering [33]. However, these space domain algorithms only perform filtering operations on grayscale images rather than color images. More importantly, these methods suffer from an extensive time complexity burden and require many auxiliary qubits. Thus, this paper proposes a spatial domain filtering algorithm for quantum color images that overcomes these problems.

This paper suggests a novel median filtering algorithm appropriate for quantum color images based on the OCQR model. The contributions of this paper are as follows:

(1) The image storage quantum circuit based on the OCQR model is optimized, enhancing multi-image representation. (2) Compared with current median filtering algorithms, the proposed algorithm presents a lower time complexity and requires much fewer auxiliary qubits. (3) The proposed algorithm is simulated on IBM Quantum Experience (IBM Q) platform, proving its feasibility.

The remainder of this paper is organized as follows: Sect. 2 discusses the storage of a color image based on the OCQR model, and Sect. 3 builds the quantum circuit of median filtering for quantum color imagery. Section 4 performs the simulation experiment on IBMQ, and finally, Sect. 5 concludes this work.

2 The storage of quantum color image

This paper adopts the OCQR [8] model. In this model, the intensity information per color channel can be easily processed, and the quantum image can be accurately restored to the classic image through quantum measurements. However, the image is prepared separately for each pixel, and therefore, it is necessary to use multi-bit-controlled NOT gates to prepare the color information throughout the entire image [8], imposing an excessive time complexity. To overcome this, we prepare the image in rows and use the information transfer module and the reset operation to convert the multi-bit-controlled NOT gates to Toffoli gates, dramatically reducing time complexity. Given that the auxiliary qubits in the information transfer module can be reused in the following image processing process, the auxiliary qubits are not added to the total qubits required in the entire algorithm. Furthermore, the reset operation can set a qubit state to zero [16]. In the following quantum circuits, the symbol $|0\rangle$ with a square outside represents the reset operation.

2.1 The improvement of quantum color image preparation module

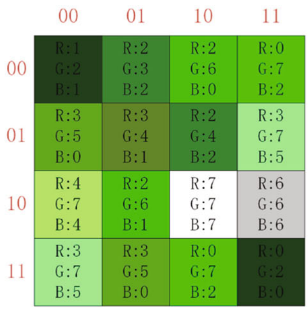
This subsection improves the color image preparation module based on the OCQR model, which encode the classic color image into the quantum superposition states described in Eq. 1. Let the size of a color image be $2^n \times 2^n$ and every channel (R,G,B) ranges in $[0, 2^q - 1]$. For the OCQR model, the color image can be written as:

$$|I\rangle = \frac{1}{\sqrt{2^{2n+1}}} \sum_{\lambda=0}^{1} \sum_{Y=0}^{2^n-1} \sum_{X=0}^{2^n-1} \otimes_{i=0}^{q-1} |C_{YX}^i\rangle |\lambda\rangle |YX\rangle \tag{1}$$

where $C_{YX}^i \in \{0, 1\}$, $i = 0, 1, ..., q-1$, which encodes the image's intensity information of the corresponding color channel, λ encodes the color channel, n qubits encode the position information on the X axis, and another n qubits encode the position information on the Y axis. Figure 1 illustrates an example of a 4×4 color image, where every (R,G,B) channel ranges in [0, 7] utilizing the OCQR model.

One pixel at position $|0000\rangle$ in Fig. 1 is selected as an example to explain the optimization process. The quantum circuit to encode the pixel to the quantum state is as illustrated in Fig. 2, and the role of each qubit is described next.

- Intensity information. In Fig. 2, from top to bottom, the first three qubits C_0 to C_2 represent the intensity information, i.e., the outputs of C_0 to C_2 are the first three qubits of binary strings 001000000, 010010000 and 001100000. The results indicate that the intensity of the RGB channel is 1, 2, and 1 in decimal, respectively.



$$
\begin{aligned}
|I\rangle = \frac{1}{8}\Big(&(|001\rangle\otimes|00\rangle+|010\rangle\otimes|01\rangle+|001\rangle\otimes|10\rangle)\otimes|0000\rangle+(|010\rangle\otimes|00\rangle+|011\rangle\otimes|01\rangle+|010\rangle\otimes|10\rangle)\otimes|0001\rangle\\
&+\left(\underbrace{|010\rangle}_{\text{Intensity}}\otimes\underbrace{|00\rangle}_{\text{Color channel}}+|110\rangle\otimes|01\rangle+|000\rangle\otimes|10\rangle\right)\otimes\underbrace{|0010\rangle}_{\text{Location information}}+(|000\rangle\otimes|00\rangle+|111\rangle\otimes|01\rangle+|010\rangle\otimes|10\rangle)\otimes|0011\rangle\\
&+(|011\rangle\otimes|00\rangle+|101\rangle\otimes|01\rangle+|000\rangle\otimes|10\rangle)\otimes|0100\rangle+(|011\rangle\otimes|01\rangle+|100\rangle\otimes|01\rangle+|001\rangle\otimes|10\rangle)\otimes|0101\rangle\\
&+(|010\rangle\otimes|00\rangle+|100\rangle\otimes|01\rangle+|010\rangle\otimes|10\rangle)\otimes|0110\rangle+(|011\rangle\otimes|00\rangle+|111\rangle\otimes|01\rangle+|101\rangle\otimes|10\rangle)\otimes|0111\rangle\\
&+(|100\rangle\otimes|00\rangle+|111\rangle\otimes|01\rangle+|100\rangle\otimes|10\rangle)\otimes|1000\rangle+(|010\rangle\otimes|00\rangle+|110\rangle\otimes|01\rangle+|001\rangle\otimes|10\rangle)\otimes|1001\rangle\\
&+(|111\rangle\otimes|00\rangle+|111\rangle\otimes|01\rangle+|111\rangle\otimes|10\rangle)\otimes|1010\rangle+(|110\rangle\otimes|00\rangle+|110\rangle\otimes|01\rangle+|110\rangle\otimes|10\rangle)\otimes|1011\rangle\\
&+(|011\rangle\otimes|00\rangle+|111\rangle\otimes|01\rangle+|101\rangle\otimes|10\rangle)\otimes|1100\rangle+(|011\rangle\otimes|00\rangle+|101\rangle\otimes|01\rangle+|000\rangle\otimes|10\rangle)\otimes|1101\rangle\\
&+(|000\rangle\otimes|00\rangle+|111\rangle\otimes|10\rangle+|010\rangle\otimes|10\rangle)\otimes|1101\rangle+(|000\rangle\otimes|00\rangle+|010\rangle\otimes|10\rangle+|000\rangle\otimes|10\rangle)\otimes|1111\rangle\Big)
\end{aligned}
$$

Fig. 1 A 4 × 4 color image represented with the OCQR model

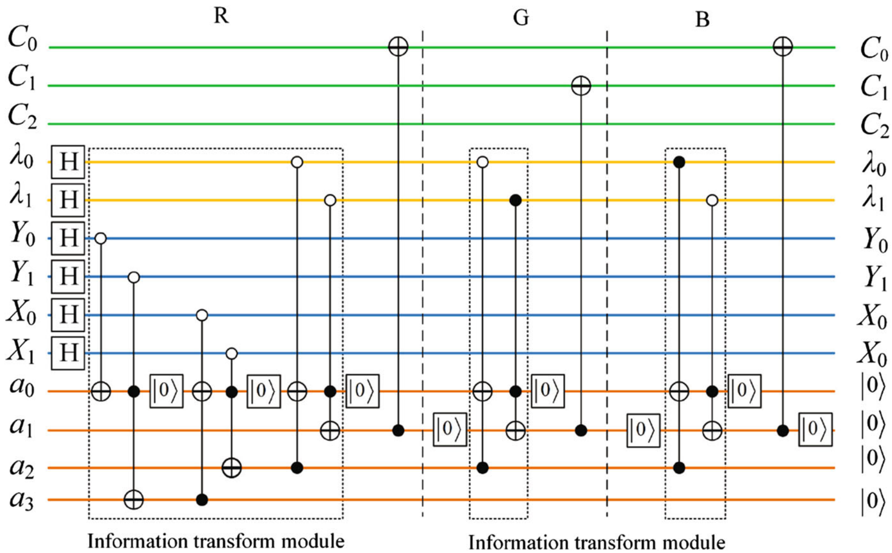


Fig. 2 Pixel preparation of the '0000' position in Fig. 1

- Color channel information. The qubits λ_0 and λ_1 represent the corresponding RGB color channel, hence $|00\rangle$, $|01\rangle$, and $|10\rangle$ indicate the 'R', 'G', and 'B' channel, respectively. The outputs of λ_0 and λ_1 are the fourth and fifth qubits of the binary strings 001000000, 010010000, and 001100000.
- Position information. The qubits Y_0 and X_0 represent the position information at the Y and X axis, corresponding to the last two qubits of the binary strings 0010011, 0100111, and 1111011 in Fig. 2.
- Auxiliary qubits. The qubits a_0 to a_3 are four auxiliary qubits that reduce time complexity and prepare the color image processed in rows.

Next, we introduce how the color image per row is prepared using the auxiliary qubits. The X,Y, and λ evolve into the quantum superposition state after H gates, which encode the position and color channel information. Step 1, the Y coordinate information of the first row in the image is transmitted to the auxiliary qubit a_3 through the quantum gates. Step 2, the position encoding of the first pixel is implemented using Toffoli gates connecting the a_3 and X coordinate information, which is transmitted to the auxiliary qubit a_2. Step 3, the location information of a_2 and the color channel information 'R' are encoded together and transmitted to the auxiliary qubit a_1. Step 4, the process adds the CNOT gate to a_1 and the intensity information C so that the position, color channel, and intensity information are in a one-by-one correspondence. Thus, the position and the color channel information can be shared, which we call an information transfer system (Fig. 2). When the intensity information of a pixel's 'R' channel is available, the auxiliary qubits a_1 and a_2 will be set to zero using the reset operation. Then, we repeat steps 3 and 4, and the position information a_2 and the color channel information 'G' and 'B' are encoded. So, the corresponding intensity information is prepared, and the position information is shared. When a pixel is prepared, a_2 is set to zero, and steps 2 to 4 are repeated to operate the first line of the other pixel values to share the Y position information. When the first row of pixels is ready, a_3 is set to zero, and the process repeats steps 1–4 to start preparing the next row of pixels. Reusing the auxiliary qubits requires only four auxiliary qubits for the entire preparation process, which do not increase as the color image size increases. The algorithm is presented in Algorithm 1.

Algorithm 1: Preparation of the quantum color image

Input: Classical color image
Output: Quantum color image
some description;
for *Input classic color image information* **do**
    for *Detect the image information of each row of the first pixel* **do**
        Y coordinate information is transmitted to the auxiliary qubit a_3;
        for *Pixels in the same row* **do**
            a_3 and X coordinate information preparation on a_2;
            for *Same pixel* **do**
                a_2 and color channel λ refer to the preparation of a_1;
                Add the CNOT gate to a_1 and intensity information C;
                Reset a_0 and a_1;
            end
        end
        Reset a_2;
    end
    Reset a_3;
end

Figure 2 highlights that the information transfer module can transmit the position and the color channel information to the auxiliary qubit through nine Toffoli gates and one CNOT gate. Then, the intensity information can be obtained through some CNOT gates instead of 4-bit control NOT gates. Although four auxiliary qubits are added for this process, these can be efficiently reused for subsequent filtering.

Let the size of a color image be $2^n \times 2^n$ and every (R,G,B) channel ranges in $[0, 2^q - 1]$. The time complexity of the storage for the color image can be calculated as follows. It requires $2n + 2$ H gates to create the position and color channel information at a quantum superposition state, and the preparation of the first-pixel value of each row in the image requires an information transmission module and $3q$ CNOT gates. An information transmission module includes $2n + 5$ Toffoli gates and a CNOT gate. $n + 6$ Toffoli gates and $3q$ CNOT gates are required to prepare each pixel from the second pixel to the end of the first row. According to [34], the time complexity of an H gate, NOT gate, and CNOT gate is one, respectively, and the time complexity of a Toffoli gate is five. Therefore, the complexity of preparing a row is as follows:

$$\begin{aligned}&(2n + 5) \times 5 + 3q + 1 + [(n + 6) \times 5 + 3q](2^n - 1) + 2n + 2\\&\quad= (5n + 3q + 30) \cdot 2^n + 7n - 2\end{aligned} \tag{2}$$

The total time complexity is as follows:

$$\begin{aligned}&\left[(5n + 3q + 30) \cdot 2^n + 5n - 4\right] \cdot 2^n + 2n + 2\\&\quad= (5n + 30 + 3q) \cdot 2^{2n} + (5n - 4) \cdot 2^n + 2n + 2\\&\quad= O((n + q)2^{2n})\end{aligned} \tag{3}$$

According to [35], an n-bit control NOT gate can be constructed by two $(n-1)$-bit control NOT gates and two Toffoli gates, which can be constructed through $(3 \times 2^{n-2} - 2)$ Toffoli gates without an auxiliary qubit. In [8], the OCQR model uses $(2n + 2)$-bit control NOT gates, and the number of the multi-bit control NOT gate is $3q2^{2n}$. Hence, the complexity is as follows:

$$\begin{aligned}&5 \times 3q \times (3 \times 2^{2n} - 2) \times 2^{2n} + q + 2n + 2\\&\quad= 45q \times 2^{4n} - 30q \times 2^{2n} + 2n + q + 2\\&\quad= O(q2^{4n})\end{aligned} \tag{4}$$

Based on [34], an n-bit control NOT gate is equivalent to $2(n-1)$ Toffoli gates and a CNOT gate, which uses $n - 1$ auxiliary qubits. If the color image encoding process of [8] adopts this strategy, the complexity reduces to:

$$\begin{aligned}&(2(2n + 1) \times 5 + 1) \times 3q \cdot 2^{2n} + 2n + q + 2\\&\quad= 60nq \cdot 2^{2n} + 33q \cdot 2^{2n} + 2n + q + 2\\&\quad= O(qn2^{2n})\end{aligned} \tag{5}$$

Nevertheless, it uses $3q \cdot 2^{2n} \cdot (2n+1)$ auxiliary qubits. Table 1 reports the comparison results considering time complexity and the number of auxiliary qubits for the color image storage process.

2.2 Preparation for neighborhood of pixels

For a filtering window of 3×3, the median filtering algorithm sorts all pixels within the window and their 8-neighborhoods and then selects the medians as outputs. The

Table 1 Image preparation compared against the original OCQR model considering auxiliary qubits and time complexity

Preparation method	Time complexity	Auxiliary qubits
[8] without auxiliary qubit	$O(q2^{4n})$	0
[8] uses auxiliary qubits	$O(qn2^{2n})$	$O(qn2^{2n})$
Our	$O((q+n)2^{2n})$	4

pixel and its 8-neighborhood preparation module will be proposed in this subsection, with the following steps comprising the neighborhood preparation module.

First, we obtain the 8-neighborhood images by performing a circular shift operation on the image to be processed before encoding them to a quantum state. As depicted in Fig. 3, the image in the center is the core image, and the surrounding eight images are the neighborhood images. For example, if we want to get the neighborhood above the central pixel, we should move all the pixels down by one pixel, as illustrated in the sub-figure of the first row and the second column in Fig. 3. It can be seen that pixels in the same position of the nine color images are the pixels to be processed and their 8-neighborhood pixels.

Second, we design a quantum state to represent the nine color images. For a $2^n \times 2^n$ color image and every (R,G,B) channel ranging in $[0, 2^q - 1]$, the position information of $2n$ qubits and the 2-qubits channel information are shared by the nine color images, defined as:

$$\begin{aligned}|I\rangle = \frac{1}{\sqrt{2^{2n+1}}}\sum_{\lambda=0}^{1}\sum_{Y=0}^{2^n-1}\sum_{X=0}^{2^n-1} |C_{Y,X}\rangle \otimes |C_{Y-1,X}\rangle \otimes |C_{Y-1,X+1}\rangle \otimes |C_{Y,X+1}\rangle \\ \otimes |C_{Y+1,X+1}\rangle \otimes |C_{Y+1,X}\rangle \otimes |C_{Y+1,X-1}\rangle \otimes |C_{Y,X-1}\rangle \otimes |C_{Y-1,X-1}\rangle \otimes |\lambda\rangle|YX\rangle\end{aligned} \tag{6}$$

where $C_{Y-1,X}, C_{Y-1,X+1}, C_{Y,X+1}, C_{Y+1,X+1}, C_{Y+1,X}, C_{Y+1,X-1}, C_{Y,X-1}, C_{Y-1,X-1}$ represent the grayscale value of eight-neighborhood pixels of $C_{Y,X}$.

Third, we design a quantum circuit that can encode the nine color images to a quantum state in Eq. 6. Since the complete quantum circuit diagram is too large, we only present the preparation of one channel information in one pixel to enhance readability. Figure 4 illustrates the preparation process of the nine color images corresponding to position information '0000' and the color channel 'R' of the nine images in Fig. 3.

Figure 4 reveals that the quantum preparation of eight images and their neighborhood can be realized by adding only several CNOT gates based on Fig. 2, which dramatically reduces the preparation complexity and the number of auxiliary qubits. The quantum circuit in the first half of Fig. 4 is part of Fig. 2, which encoding the position information '0000' and color channel 'R' of all images in Fig. 3. The subsequent CNOT gates encode the pixel values of the color channel 'R' at position '0000'. For example, the pixel value at position '0000' of the first picture in Fig. 3 (Position information is $Y+1$, $X-1$) is 0, and there is no need to add additional CNOT gates, as its default value is 0. The pixel value at position '0000' of the second image (Position information is $Y+1$, X) is 3; then, it needs to use CNOT gates to make the value of

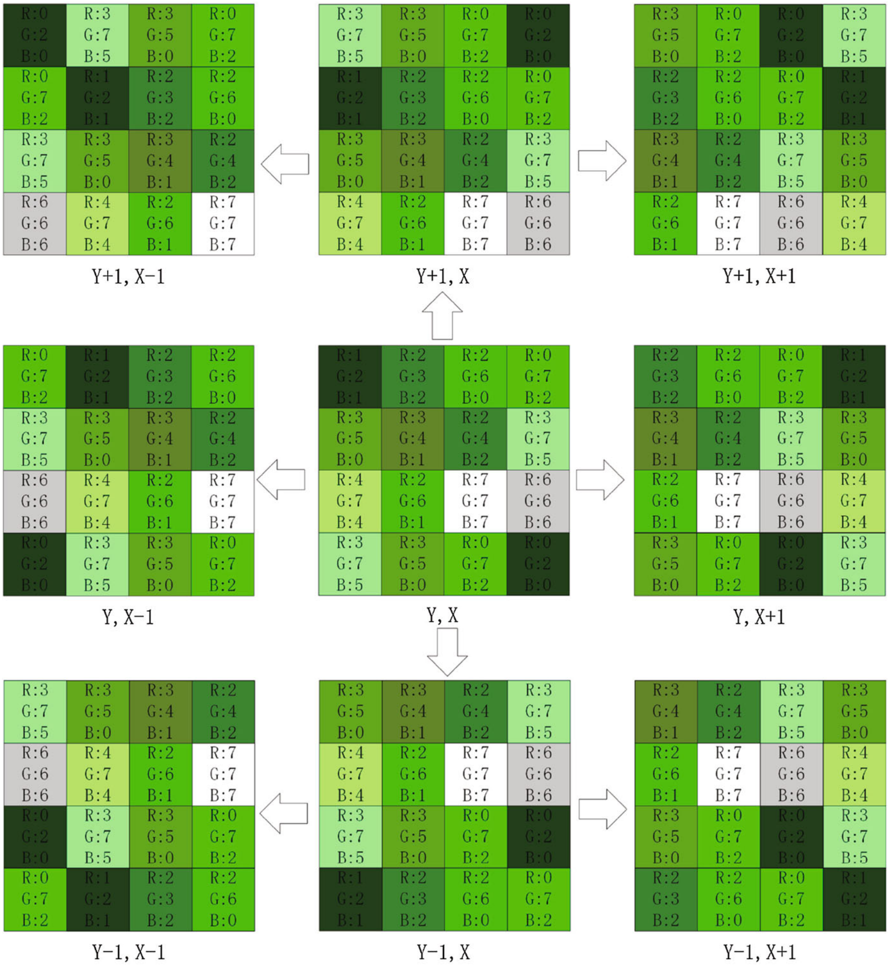


Fig. 3 4×4 color image and its eight-neighborhood images

$C^1_{Y+1,X}$ and $C^2_{Y+1,X}$ equal to 1, and so on, we can complete the preparation of other images. Table 2 reports the comparative results considering the time complexity for preparing a color image and its 8-neighborhood against [8, 30, 31].

3 Median filtering algorithm of quantum color image

This section designs the median filtering algorithm for quantum color imagery. First, we introduce the comparator [36], controlled switch [33], and sorting [31] modules. Then, we design the median filtering algorithm for the quantum color images utilizing these modules.

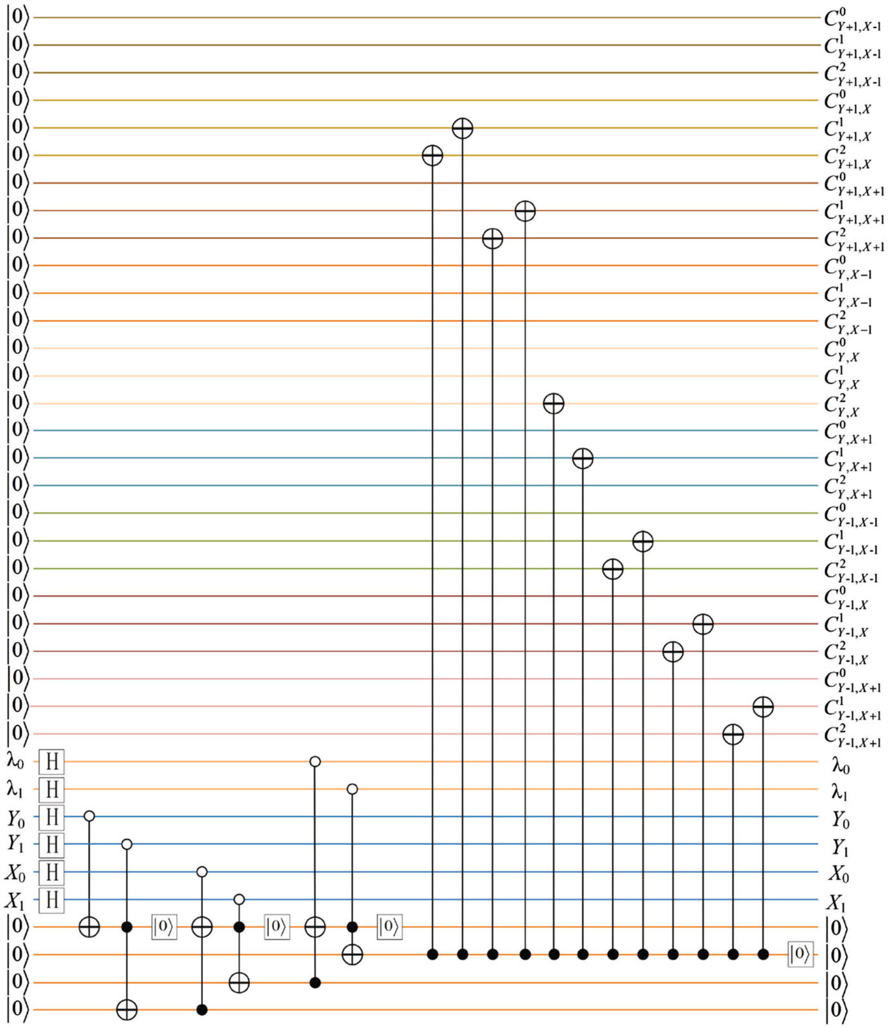


Fig. 4 Preparation of nine quantum color images corresponding to position information "0000" and color channel 'R' in Fig. 3

Table 2 Quantum color images and their eight-neighborhood preparation challenged against other methods in terms of auxiliary qubits and time complexity

Neighborhood representation	Time complexity	Auxiliary qubits	Quantum color image	Simulate
[30]	$O(qn2^{2n} + 28n^2)$	$O(qn2^{2n})$	No	No
[31]	$O(qn2^{2n} + 10n^2)$	$O(qn2^{2n})$	No	No
[8]	$O(qn2^{2n})$	$O(qn2^{2n})$	Yes	No
Our	$O((q+n)2^{2n})$	4	Yes	Yes

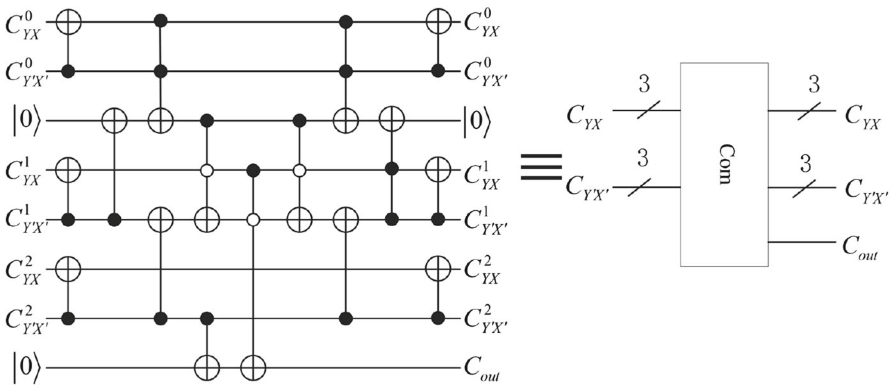


Fig. 5 Three-digit quantum comparator

3.1 Comparator module

Suppose there are two numbers a and b. Although current comparators [30, 31] can distinguish three states, i.e., $a > b$, $a < b$, and $a = b$, this paper adopts the quantum comparator of [36], which can distinguish only two states $a > b$ and $a \leq b$, but affords a lower time complexity.

Suppose a composite system comprises two n-qubits quantum states $|C_{YX}\rangle|C_{Y'X'}\rangle$. The quantum comparator compares the qubit strings $|C_{YX}\rangle = |C^{n-1}_{YX}C^{n-2}_{YX}\cdots C^0_{YX}\rangle$ and $|C_{Y'X'}\rangle = |C^{n-1}_{Y'X'}C^{n-2}_{Y'X'}\cdots C^0_{Y'X'}\rangle$. $|C_{\text{out}}\rangle$ is the output qubit of the quantum comparator. When $|C_{YX}\rangle < |C_{Y'X'}\rangle$, $|C_{\text{out}}\rangle = |1\rangle$. When $|C_{YX}\rangle \geq |C_{Y'X'}\rangle$, $|C_{\text{out}}\rangle = |0\rangle$. It is worth noting that the auxiliary qubits do not increase as the numbers involved in the comparison increase. Compared with [30, 31], the time complexity of the proposed quantum comparator decreases from an exponential level to a polynomial level. The corresponding quantum circuit diagram is illustrated in Fig. 5.

3.2 Swap module

A quantum swap module aims to realize the exchange of two pixels [33], with Fig. 6 presenting the quantum circuit of a swap module. The latter figure reveals that the swap module comprises some controlled swap gates and the control bit is the result qubit $|C_{\text{out}}\rangle$ of the comparator. For example, given a two n-bit composite system $|C_{YX}\rangle|C_{Y'X'}\rangle$, among them, $|C_{YX}\rangle = |C^{n-1}_{YX}C^{n-2}_{YX}\cdots C^0_{YX}\rangle$, $|C_{Y'X'}\rangle = |C^{n-1}_{Y'X'}C^{n-2}_{Y'X'}\cdots C^0_{Y'X'}\rangle$. When $C_{\text{out}} = 1$, $|C_{YX}\rangle|C_{Y'X'}\rangle$ transforms into $|C_{Y'X'}\rangle|C_{YX}\rangle$ after passing through the quantum swap module. When $C_{\text{out}} = 0$, no transformation occurs. Compared with [30, 31], the output qubit of the swap module is reduced to one qubit, which will significantly reduce time complexity.

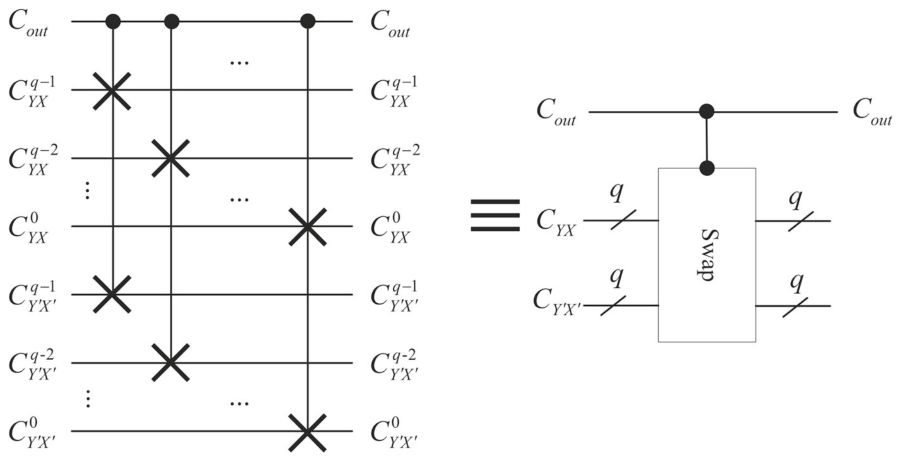


Fig. 6 Quantum controlled swap module

3.3 Sort module

The sorting module comprises three comparison modules and three controlled swap modules and is designed to sort three pixels in ascending order. This module sorts three intensity values to calculate the maximum, intermediate, and minimum. Unlike [31], in this paper, the output qubit of the comparator module is reused by the reset operation. Thus, the total number of qubits required in the sort module is reduced. As shown in Fig. 7, when three pixel values $|C_{Y,X}\rangle = |C_{Y,X}^{q-1}C_{Y,X}^{q-2}\cdots C_{Y,X}^{0}\rangle$, $|C_{Y+1,X}\rangle = |C_{Y+1,X}^{q-1}C_{Y+1,X}^{q-2}\cdots C_{Y+1,X}^{0}\rangle$, $|C_{Y+1,X+1}\rangle = |C_{Y+1,X+1}^{q-1}C_{Y+1,X+1}^{q-2}\cdots C_{Y+1,X+1}^{0}\rangle$ are inputs, the outputs are $|C_{Y,X}^{'}\rangle = Max$, $|C_{Y+1,X}^{'}\rangle = Med$, and $|C_{Y+1,X+1}^{'}\rangle = Min$. Thus, this sort module can realize the sort function, utilizing a single auxiliary qubit.

3.4 Median filtering algorithm of quantum color image

Next, we design the median filtering algorithm for a quantum color image. We adopt the method of [31] to obtain the median, involving the following steps. First, we sort the three elements of each column in the filter window (Fig. 8a), where $B7$, $B8$ and $B9$ are the maximum values of each column (Fig. 8b). Then, the three pixels of each row are sorted, and $C3$, $C6$, and $C9$ are the maximum values of each row (Fig. 8c). Finally, we sort the diagonal three pixels $C3$, $C5$, and $C7$, and the median value $D2$ (Fig. 8d) of the diagonal three pixels is the median value of the nine pixels in the filter window. For the detailed proof, the reader is referred to [31].

Due to quantum parallelism, when median filtering is performed on one pixel, it is also performed on the other pixels. Figure 9 illustrates the quantum circuit that realizes the median filtering algorithm for quantum color images. Table 3 reports the comparison results among existing median filtering algorithms of quantum grayscale images and the proposed algorithm, where q represents a total of 2^q intensity levels. The time complexity of a q-bit control NOT gate is $10(q-1)+1 = 10q-9$. It should

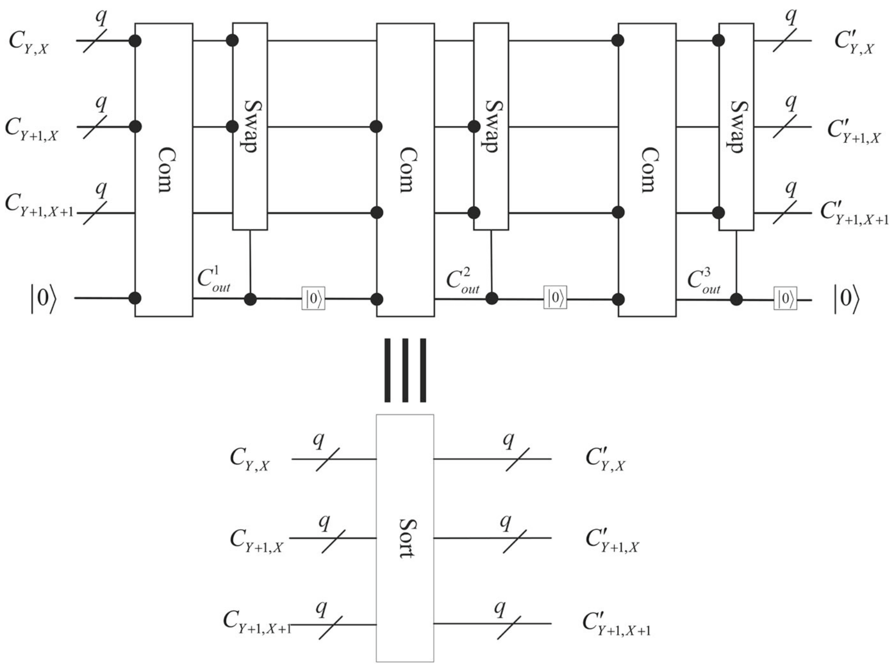


Fig. 7 Sort module

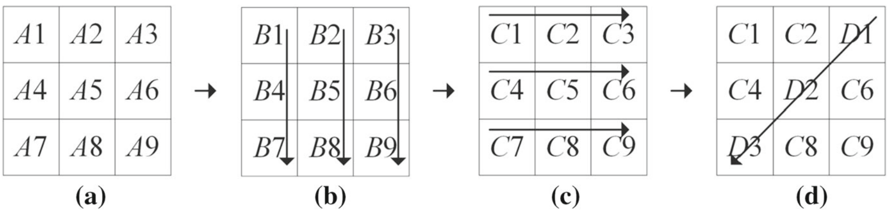


Fig. 8 Median calculation [31]

be noted that [31] calculates only the time complexity of the swap gate and ignores the time complexity of the t-bit control swap gate. Furthermore, Jiang et al. [31] uses a 2-bit control swap gate, namely three 3-bit control NOT gates, so the time complexity of a 2-bit control swap gate is $21 \times 3q = 63q$. Nevertheless, the proposed algorithm adopts the 1-bit control swap gate, which includes three Toffoli gates, and therefore, the time complexity of a 1-bit control swap gate is $5 \times 3q = 15q$.

In conclusion, the median filtering algorithm for quantum grayscale images [31] adopts 21 comparators (whose time complexity is $21q^2$) and 21 2-bit control swap gates, with a time complexity of$21q^2 + 63q \times 21 = 21q^2 + 1323q$. The comparator used in this article has a time complexity of $14q - 6$, and a total of 21 comparators (time complexity of $21 \times (14q - 6) = 294q - 126$) and 21 control swap gates (time complexity of $21 \times 15q = 315q$). Thus, the total time complexity of the proposed quantum median filtering algorithm is $609q - 126$. In addition, we use the reset

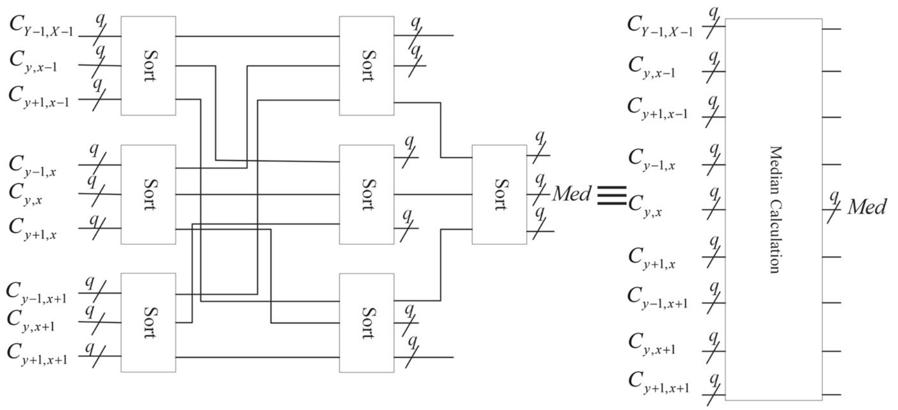


Fig. 9 The quantum circuit of the median filtering algorithm

Table 3 Advantages and disadvantages of the existing median filtering algorithm in time complexity, auxiliary qubits, and simulation

Median filtering algorithms	Time complexity	Auxiliary qubit numbers	Simulation
[30]	$30q^2 + 1890q$	$O(q)$	No
[31]	$21q^2 + 1323q$	$O(q)$	No
Our	$609q - 126$	2	Yes

operation to reuse qubits, and thus, the total number of auxiliary qubits during median calculation is constant.

4 The simulation of the proposed algorithm on IBM Q platform

This section simulates the proposed algorithm on the IBM Q platform. IBM has several real quantum devices and simulators available for users through the cloud. These devices are accessible and can be used through Qiskit, an open-source quantum software development kit, and IBM Q Experience, which offers a virtual interface for coding a quantum computer. In addition, researchers can use a local simulator through Qiskit. In the Qiskit framework, the quantum devices, quantum simulator, and local simulator are all called backend. As of May 2022, 5000 qubits can be achieved through the backend ('Simulator_stabilizer'). The corresponding quantum circuit for median filtering of a quantum color image can be simulated on the IBM Q platform, and the overall quantum circuit can be verified by the output data, reflected by the probability histogram of the quantum circuit. For further details on the IBM Q platform, the reader is referred to our previous work [16].

Due to the 5000 qubits provided by the IBM Q platform, we can simulate big-sized color images. Considering that the output probability histogram is relatively large, this paper simulates the spacial median filtering algorithm of the 4×4 color image with intensity values ranging from 0 to 7. The median position *med*, color channel

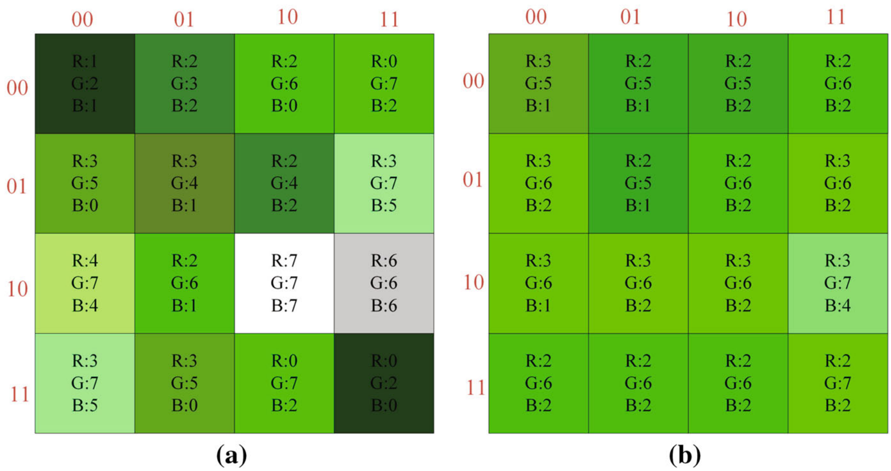


Fig. 10 **a** the color image to be filtered and **b** a filtered color image

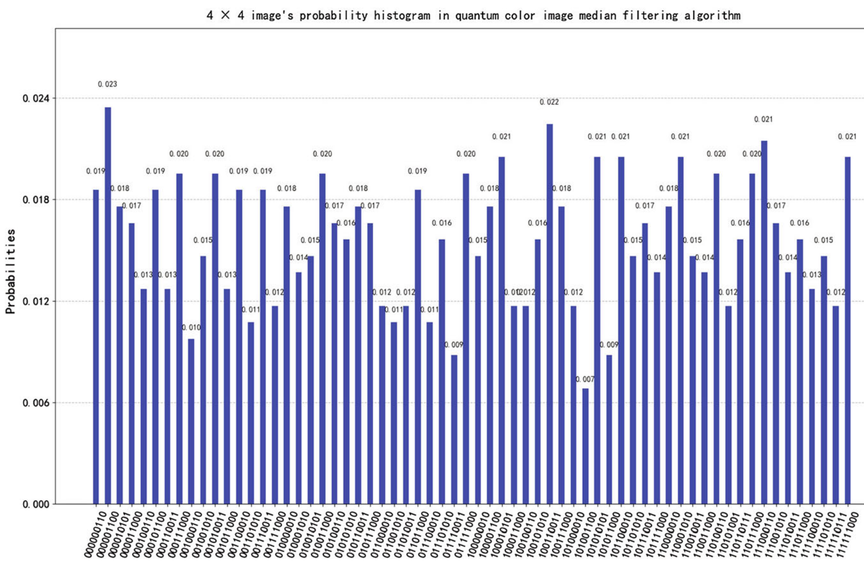
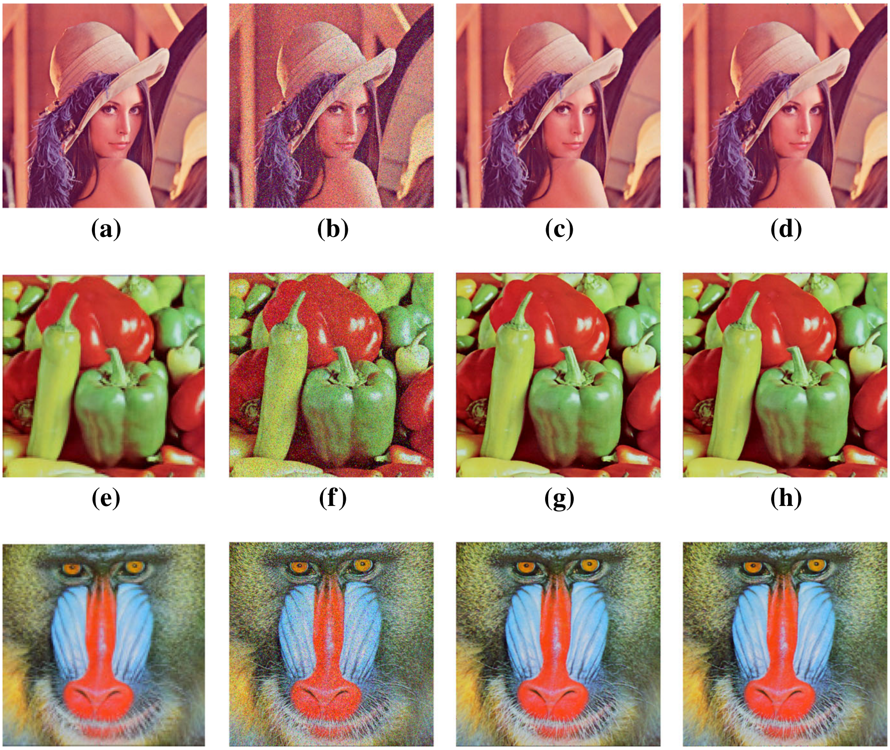


Fig. 11 4 × 4 image probability histogram after filtering

λ, and position information YX are measured after median filtering, and the number of quantum measurements is 256. Figure 10a presents the color image to be filtered, Fig. 11 depicts the probability histogram of the color image after filtering, and Fig. 10b is the filtered color image.

The strings in abscissa of Fig. 11 are the quantum state outputs of the quantum measurements. For example, considering the first string '000000110', from right to left, the first four qubits are the position information of the color image, the fifth and sixth qubits are the color channel information, and the last three qubits are the



(a) (b) (c) (d)

(e) (f) (g) (h)

(i) (j) (k) (l)

Fig. 12 Denoising performance of quantum color image median filter and classical color image filter. **a**, **e**, **i** The original images. **b**, **f**, **j** Salt-and-pepper noise images with a density of 0.1. **c**, **g**, **k** The result of median filtering of classical color image. **d**, **h**, **i** The filtering results using the proposed scheme

intensity value information. The vertical axis shows the probability of the quantum state under quantum measurements. According to Fig. 11, the filtered classical color image Fig. 10b can be obtained, revealing that noise at coordinates (0,0), (2,2), (2,3) and (3,3) is filtered out.

In addition, three standard test color images of size 512×512, such as Lena, Peppers, and Baboon, are chosen. We add to these images 0.1 salt and pepper noise and use them as the input of the color median filter circuit. Each color image is measured 3.2 million times. The maximum number of measurements on the IBMQ is 20,000, so we repeat the entire circuit 16 times to get the classic filtered image information. The final result of the whole circuit has 1,048,576 strings, so we cannot show its probability histogram. According to Fig. 12, we find that the filtering results are the same as in the median filtering algorithm for classic color images, and our algorithm's complexity $O(q)$ is far less than the classic color image median filtering algorithm $O(2^{2n})$.

5 Conclusion

This paper designs a novel and efficient spatial median filtering algorithm for quantum color images based on the OCQR representation mode, simulated on the IBM Q platform. Specifically, this work first presents a novel method of preparing quantum images based on the OCQR model. The images are encoded in rows, and the information transfer module is designed to convert the multi-bit-controlled NOT gates to Toffoli gates, which dramatically reduces time complexity. Second, we develop a novel method to obtain the neighborhood pixels. Eight color images are used to store neighborhood information, and their position and color channel qubits are shared with the core image, fully exploiting quantum parallelism and thus improving the performance of the subsequent process. Third, we suggest a spatial median filtering algorithm for quantum color imagery based on basic modules. The advantages of the suggested median filtering algorithm are lower time complexity and fewer auxiliary qubits. Compared with current algorithms, the proposed median filtering algorithm affords a reduced time complexity from $O(q^2)$ to $O(q)$, and the number of auxiliary qubits reduces from $O(q)$ to $O(1)$. The quantum median filtering algorithm provides exponential speed-up compared with its classic counterpart. In addition, the proposed algorithm is simulated in the IBM Q quantum simulator through the Qiskit framework, demonstrating the feasibility of the proposed algorithm.

The multiple image encoding and processing methods are applicable for image neighborhoods in single image processing schemes and can be applied to quantum machine learning that needs to process large amounts of image data. Nevertheless, more interesting practical applications involving quantum image processing and machine learning should be explored. Overall, the idea discussed in this paper shall provide an efficient solution for many problems in the big data era.

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